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New Early Devonian paleomagnetic data from NW France: Paleogeography and implications for the Armorican microplate hypothesis

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Abstract. To constrain the Early Devonian paleogeography of the Armorican Massif, and determine the tectonic relationships between the Armorican and Bohemian Massifs (the Armorican microplate hypothesis), a paleomagnetic study of Lochkovian/Pragian sediments from the Pointe de l'Armorique, Brittany, NW France has been carried out. After removal of a present-day overprint, a higher unblocking temperature component of magnetization is identified (the B direction) which yields an overall mean direction of 222°/+35°, k=121.6, α_{95}=4.2° after bedding correction, 11 sites. No fold test is possible, but the presence of a thin sill allowed sampling for a contact test, the results of which are positive. Thus the B direction of magnetization is considered to be primary in origin and closely bracketed with the rock age. The resulting paleopole (315°E, 13°S) corresponds to a paleolatitude of 19°S for the Armorican Massif, placing it adjacent to the southern margin of Baltica and Avalonia. This clearly demonstrates for the first time that this sector of the Rheic Ocean had closed by Early Devonian times, and that the Armorican Massif was in its present-day orientation with respect to Baltica. These results are significantly different to existing paleomagnetic data from Bohemia which indicate large scale (~140°) rotation between the earliest and latest Devonian, and thus provide important information for the pre- and syncollisional tectonic relationships of these Massifs. As it is unlikely that such large scale differential crustal rotations could have occurred in an intraplate setting, it is proposed that the Armorican and Bohemian Massifs did not form a single coherent microplate, but belonged to a fairly loose mosaic of terranes prior to collision with Avalonia. Thus, the concept of an Armorican microplate is no longer tenable, and the term Armorican terrane assemblage (ATA) is proposed in reference to the Gondwana derived Paleozoic terranes now situated within the European Variscan fold belt south of the Rhenohercynian Zone (Avalonia).

1. Introduction

For early Paleozoic times, the drift histories of most of the Iapetus-bordering continents are now fairly well constrained. It is clear that by the Early Devonian the Iapetus Ocean separating Baltica and Laurentia, and the Tornquist Ocean between Baltica and Avalonia had both closed [*Torsvik et al.*, 1992, 1993, *MacNiocall and Smethurst*, 1994, *Cocks et al.*, 1997]. From paleomagnetic evidence, it is clear that Laurentia had rotated slightly from its equatorial position in Ordovician and Silurian times and, in the Early Devonian, was situated between paleolatitudes of approximately 5°N and 45°S [*MacNiocall and Smethurst*, 1994]. Baltica was straddling the equator in the Early Devonian [*Torsvik et al.*, 1992], and the southern margin of eastern Avalonia was at about 20°S [*Channell et el.*, 1992]. The position of Gondwana is more controversial, and basically two different models have been proposed. Discussion of these is outside the scope of this paper (see *Bachtadse et al.* [1995] and *Van der Voo* [1993] for details); the model preferred in this paper is the simpler one, which places the northern margin of Gondwana at about 60°S in the Early Devonian. The situation for the so-called Armorican microplate, generally considered to comprise the Iberian, Armorican, and Bohemian Massifs, and the Rheic Ocean separating it from the southern margin of Avalonia, is also unclear [*Van der Voo*, 1993].

Geological evidence suggests that the "Armorican Microplate" was in fact a tectonic collage of (semi)autonomous terranes [*Franke et al.*, 1995] which were separated from Avalonia by the Rheic Ocean in the early Paleozoic. Correlation between these terranes for early Paleozoic times, however, is difficult on account of the general breakdown of faunal communities and endemism in the transgression following the Late Ordovician glaciation [*Brenchley*, 1984]. Unfortunately, the paleomagnetic data for this time period are sparse and provide conflicting scenarios. Although it is now known that Bohemia started drifting northwards away from Gondwana in the mid-Ordovician [*Tait et al.*, 1995], it is not clear whether the Armorican and Iberian Massifs also began to drift northward [*Perroud et al.*, 1983], if they remained at high paleolatitudes until at least the Late Ordovician and then rifted from Gondwana [*Perroud and Van der Voo*, 1985, *Perroud et al.*, 1986, *Cogné*, 1988], or even if they remained adjacent to the northern margin of Gondwana throughout the Paleozoic [*Cocks and Fortey*, 1990].

By Late Silurian/Early Devonian times geological evidence and paleomagnetic data indicate that in central and eastern Europe the Rheic Ocean between the Bohemian Massif and the southern

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margin of eastern Avalonia had closed [*Franke and Onken*, 1995, *Tait et al.*, 1994]. Final consolidation, however, did not occur until sometime later as evidenced in the geological record, which shows a reopening of the Rheic suture in Middle Devonian times and formation of the relatively narrow Rhenohercynian Ocean [*Franke and Onken*, 1995]. Also, the paleomagnetic record demonstrates large scale rotation of the Bohemian Massif between Siluro-Devonian and Late Devonian times [*Tait et al.*, 1997]. It is not clear, however, if these rotations are to be extrapolated to the Armorican and Iberian Massifs, or indeed if these Massifs were even situated at similar paleolatitudes. Biostratigraphic and paleoecological studies of uppermost Silurian sequences from the Armorican and Bohemian Massifs show a direct correlation between the ostracod faunas of these two areas [*Kriz and Paris*, 1982], suggesting similar paleolatitudes and no major oceanic separation. This, however, has yet to be demonstrated by paleomagnetic evidence. The only paleomagnetic results available for the Siluro-Devonian of the Iberian and Armorican Massifs are from the San Pedro red beds in Asturia, Spain [*Perroud and Bonhommet*, 1984], and the Almaden, southern Spain [*Perroud et al.*,1991], both of which have subsequently been shown to be remagnetized [*Parés and Van der Voo*, 1992, *Van der Voo et al.*, 1997].

The only well-constrained paleomagnetic data available for the Devonian of central Europe are mid-Upper Devonian results from the Vosges and Black Forest [*Bachtadse et al.*, 1983]. The data from these two regions are in good agreement and indicate paleolatitudes of approximately 15°S for central Europe. This is in accordance with the geological record that by the Late Devonian there was no longer any major oceanic separation between the Paleozoic terranes of central Europe. They also demonstrate that the mid-Late Devonian Rhenohercynian Ocean was either purely of a longitudinal nature, or its latitudinal extent was within the error of paleomagnetic data. In contrast, however, paleomagnetic data from metamorphic rocks in the Massif Central, France, suggest paleolatitudes of 39°S in the Late Devonian [*Edel*, 1987], implying that this part of Armorica remained adjacent to the northern margin of Gondwana and did not collide with the northern continents until late Carboniferous times. These data, however, are rather poorly defined, and the paleogeographic setting for this part of Armorica, as well as the timing of suturing with Avalonia, remains unclear. For Carboniferous times, paleomagnetic data demonstrate that there were significant amounts of internal rotation, oroclinal bending, and deformation within the Variscan fold belt [*Bachtadse and Van der Voo*, 1986; *Perroud*, 1986; *Tait et al.*, 1996; *Van der Voo et al.*, 1997]. This deformation clearly postdates the rotation within the Bohemian Massif which occurred prior to the Famennian, and is most likely related to late stage compression of the orogen and the collision of Gondwana to the south.

In summary, the dynamics involved and interaction of the various units consituting Armorica in Siluro-Devonian times and timing of closure of the more easterly segment of the Rheic Ocean between the Armorican Massif and eastern Avalonia remain uncertain. In order to help resolve this problem, and to constrain the position of the Armorican Massif and its tectonic affinities in the Early Devonian, a detailed paleomagnetic study of Lochkovian to Pragian age carbonates from the Pointe de l'Armorique, Brittany has been undertaken.

2. Geological Setting

The Armorican Massif (Figure 1) can be divided into three domains separated by the North and South Armorican shear zones (NASZ and SASZ). These two major deep-seated fractures served as ductile transtensional shear zones during the early tangential phases and later served as ductile, then brittle, extensional shear zones during the late Variscan periods [*Peucat et al.*, 1984; *Audren*, 1987]. The NASZ marks the boundary between the stable Cadomian blocks of the North Armorican Composite Terrane (NACT) and the Central Armorican Domain, both of which were deformed during the Variscan orogeny. In the NACT the plutonic and sedimentary complexes were metamorphosed during the Caledonian and Variscan orogenic phases (440-385 Ma). Granodiorites of calc-alkaline affinity and Siluro-Ordovician age eclogites record a subduction zone environment during the Ordovician [*Paquette et al.*, 1987; *Le Corre et al.*, 1989]. The SASZ is also a major crustal feature, including thrust-bound slices of metasedimentary and igneous rocks. Metamorphism to the south of the SASZ reached eclogite facies, with development of blueschists and a high-temperature, low-pressure migmatite belt [*Pickering and Smith*, 1995]. These have been interpreted as remnants of an accretionary complex above a subduction zone active during the Silurian [*Peucat and Cogné*, 1977]. Thus while similar series of events have been documented in both central and southern Armorica [*Jones*, 1991], the Central Armorican Domain is dominated by Variscan age dextral strike-slip transtension and is much less deformed and less metamorphosed than the adjacent regions [*Matte*, 1991].

An Early Devonian (400-385 Ma) medium pressure metamorphic event in the NACT [*Jones*, 1994], which culminated in anatexis, is clearly related to the melting of continental crust and is thought to mark the onset of continental collision [*Matte*, 1991; *Jones* 1994]. Later strike-slip tectonics and granite intrusions characterized an intracontinental environment during the Carboniferous [*Le Corre et al.*, 1989].

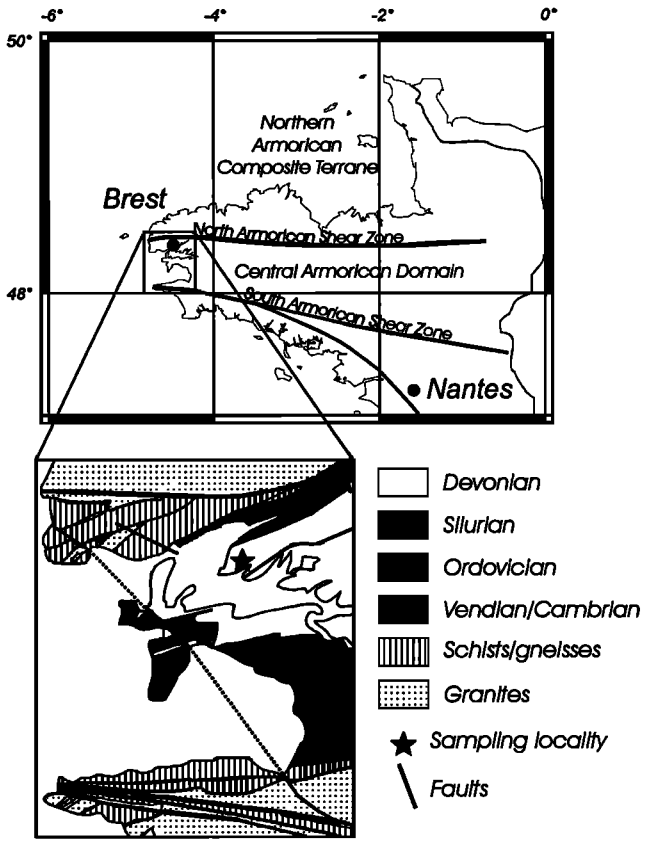


Figure 1. Geographical and geological sketch maps of northeastern France and the sampling locality.

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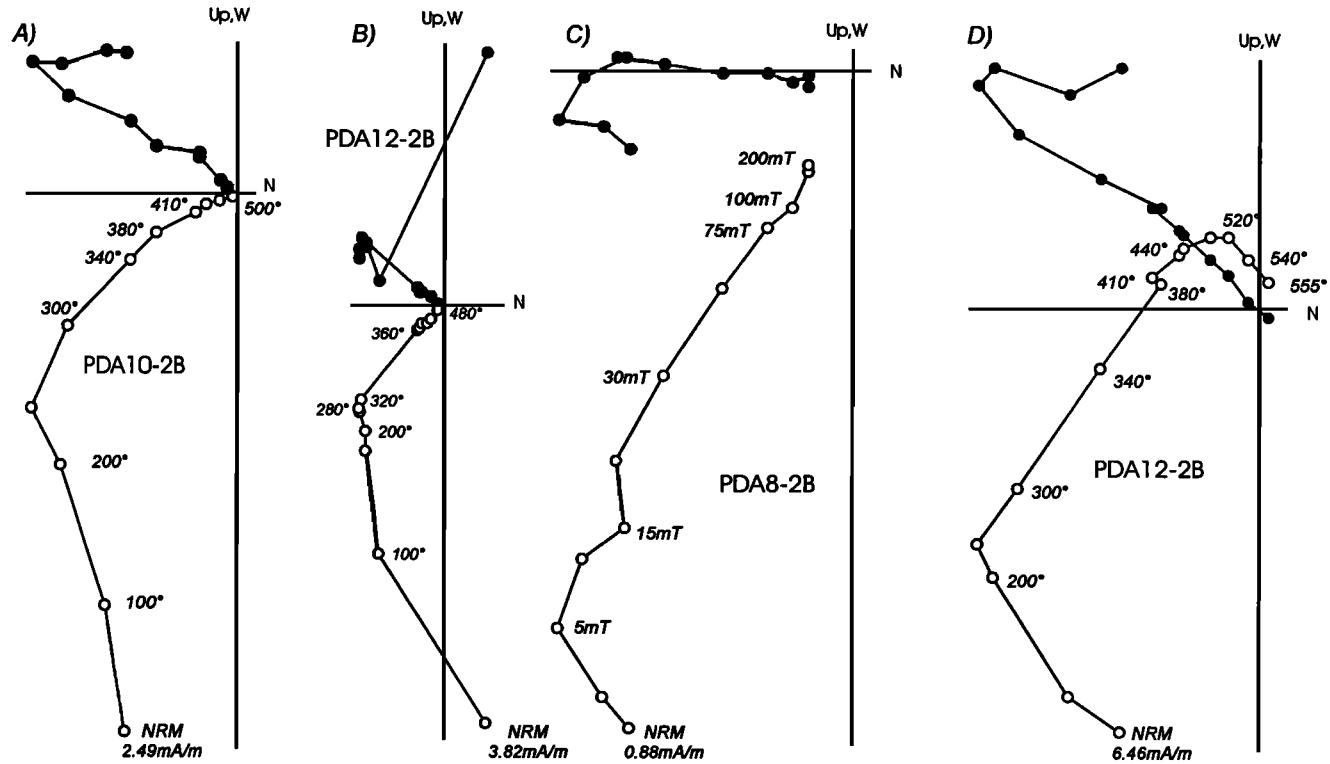


Figure 2. Thermal and alternating field demagnetization characteristics. All directions are plotted in geographical coordinates, and solid (open) symbols represent the horizontal (vertical) component. These examples show removal of component `A´ between 20° and 250°C (or up to 5 mT), followed by removal of component `B´ at higher temperatures and fields either (a-c) as a stable end point direction or (d) as a great circle trajectory (2D).

Samples for the present study were collected from essentially undeformed and unmetamorphosed Lower Devonian sequences of the Pointe de l´Armorique near Brest in Brittany. Although many Paleozoic rocks in this region have undergone fairly severe deformation and despite the relative proximity of the NASZ, these sediments have been only very gently tilted with no penetrative deformation as demonstrated by well-preserved and undeformed fossils. These rocks, which constitute a complete sequence from the uppermost part of the Grès de Landévannec (mid-Lochkovian) and the Armorican Carbonates which straddle the Lochkovian-Pragian boundary, represent the earliest development of a carbonate reef complex in the Armorican Massif [*Burchette* 1981].

3. Laboratory Procedures

Field-drilled cores were cut into standard 2.2 x 2.5 cm specimens, and magnetizations were measured using a 2G-Enterprises magnetometer in a magnetically shielded room at the University of Munich. Samples were demagnetized using both stepwise thermal and alternating field (AF) techniques, with measurement of the susceptibility of magnetization using a KLY2 Kappameter after each heating step to monitor thermochemical alteration.

Demagnetization results were analyzed using orthogonal vector plots [*Zijderveld*, 1967] and stereographic projection. Demagnetization trajectories were calculated using principal component analysis [*Kirschvink*, 1980], and great circle analysis was performed using the method of *McFadden and McElhinny* [1988]. Various rock magnetic experiments were also carried out to characterize the carriers of magnetization. These include isothermal remanent magnetization (IRM) measurements, AF and thermal demagnetization of the saturation IRM (SIRM), thermomagnetic analyses, and determination of the hysteresis properties.

4. Paleomagnetic Results

During the demagnetization experiments, two directions of magnetization can be clearly identified. The first (component A), identified in samples from most sites, is generally removed below 240°C (Figure 2). The direction of this component corresponds to that of the local present day Earth´s magnetic field and is thus considered to be of recent origin. Occasionally, there is a marked decrease in intensity after heating to 150°, indicating the presence of goethite. A second direction of magnetization (component B) is then removed at higher temperatures. In many samples there is a marked decrease in intensity at 300°-350° (Figure 2b), and thermochemical alteration at 450°-500°C occasionally obscures the ancient direction of magnetization. This inflection and subsequent alteration are probably due to the presence of pyrrhotite. Nevertheless, the southwesterly and down B component of magnetization (Table 1) is generally directed toward the origin in orthogonal projection of results (Figures 2a to 2c). In sites where no thermochemical alteration is observed at higher temperatures, the B component persists as the stable end point direction until complete demagnetization between 450° and 550°C. Alternating field demagnetization of these samples up to 250 mT reduces the magnetization to $<5\%$ of its original intensity, indicating a relatively low coercivity mineral, probably magnetite, as the principal carrier of the B magnetization (Figure 2c).

Occasionally, however, the demagnetization trajectory is not directed toward the origin in orthogonal projection (Figure 2d), indicating the presence of a component with higher unblocking temperatures. Stereographic projection demonstrates that the demagnetization trajectories move along great circle paths from the southwestern, toward the northeastern quadrant with negative

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Table 1. Site Mean Directions for Component B

Site	*N/N*(n)*	In Situ			Bedding Corrected			Bedding
		Dec/Inc, deg	*k*	α_{95}, deg	Dec/Inc, deg	*k*	α_{95}, deg	
PDA1	7/7	208/+42	65.4	7.5	220/+38	65.4	7.5	288/14
PDA2	4/5	210/+38	156.5	7.4	222/+37	156.5	7.4	300/15
PDA3	6/6	209/+46	76.3	7.7	217/+42	76.3	7.7	275/09
PDA4	5/7	214/+42	873.1	2.6	219/+38	873.1	2.6	275/09
PDA5	6/6	216/+38	94.9	6.9	231/+31	94.9	6.9	295/22
PDA6	6/6	212/+40	23.2	14.1	224/+38	23.2	14.1	300/15
PDA7	6/6	213/+39	33.3	11.8	226/+35	33.3	11.8	299/18
PDA10	4/5	207/+28	171.1	7.0	214/+27	171.1	7.0	296/14
PDA11	5/5(3)	033/+42	99.6	8.8	042/+34	99.6	8.8	275/14
PDA12	7/7(7)	033/+33	196.3	5.3	037/+34	196.3	5.3	263/08
PDA15	8/8(7)	045/+53	87.4	6.7	059/+36	87.4	6.7	270/22
Mean	11 sites	212/+40	121.7	4.2	222/+35	121.6	4.2	
					Pole Position 315°E; 13°S			
		Contact Test						
PDA13	6/6	211/+29	852.4	2.3	215/+20	1048.7	2.1	266/14
								254/14
PDA14	6/6	210/+22	264.3	4.1	214/+14	264.3	4.1	271/15
mean 12 samples		210/+26	222.5	2.9	215/+17	284.6	2.6	

N, number of samples measured; *N**, number of samples used in calculation of site mean; (*n*), number of great circles; Dec/Inc, declination/inclination (in degrees); *k*, precision parameter [after *Fisher* 1953]; α_{95}, semiangle of the cone of 95% confidence. Bedding correction is given in terms of azimuth/dip.

inclination values (Figure 3), i.e., toward the inverse direction of component B. This behavior is observed in samples from three sites, from the lower (older) part of the sequence studied, and is thought to indicate the superposition of the antipodal B direction. Assuming this to be the case, the method of *McFadden and McElhinny* [1988] was used to analyze the great circle data (Table 1, Figure 4). Although the great circles are all rather parallel (being anchored presumably by the southwesterly B component) and no stable end point directions could be unambiguously identified, this inverse direction is considered significant given the fact that the great circle behavior is stratigraphically controlled, and from the behavior shown in Figures 3 and 4 where the great circle paths clearly move past the southwesterly cluster of stable end point directions and into the northeastern quadrant.

Samples were also collected from a thin sill (site PDA 13) intruded into the carbonate sequence and the immediately adjacent baked sediments (site PDA 14). Although no radiometric ages are available for the sill, on geological grounds it is thought to have been emplaced only shortly after consolidation of the Lower Devonian sediments, i.e., in late Devonian/early Carboniferous times. During thermal demagnetization, samples from these two sites yield well defined components of magnetization between approximately 300°C and 580°C (Figure 5). The directions identified are similar to the B component of magnetization, but with slightly shallower inclination values (Table 1, Figure 5). The mean direction obtained from 12 samples is significantly different from the overall mean direction obtained for the B component, both in situ and after bedding correction, and thus indicates that the sequence has not been remagnetized, at least since emplacement of the sill.

5. Rock Magnetic Results

During IRM experiments saturation is generally reached by 350-400 mT, demonstrating the predominance of low and intermediate coercivity minerals as the remanence carriers (Figure 6). Occasionally, there is a very slight but gradual increase in magnetization at higher fields, up to the maximum applied field of 1.5 T. Of interest to note is that samples from sites which showed great circle behavior tend to have higher values of saturation of

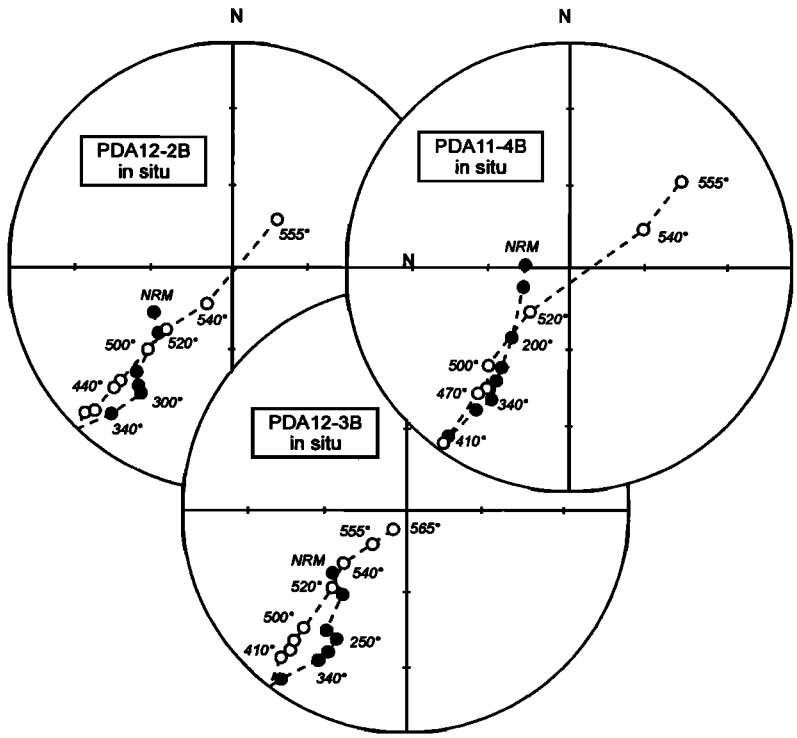


Figure 3. In situ equal area stereographic projection of samples showing great circle behaviour. Solid (open) symbols represent lower (upper) hemisphere projection.

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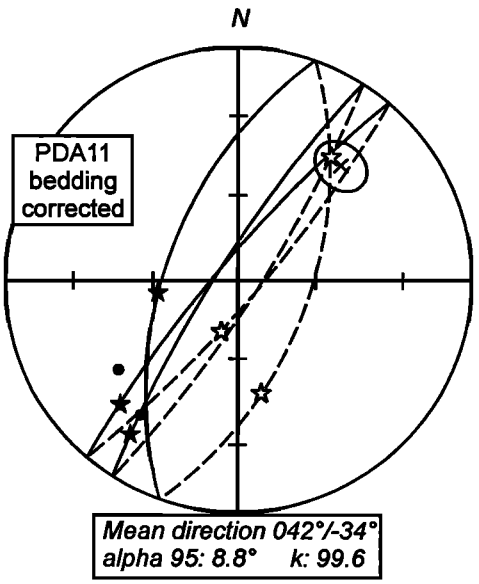


Figure 4. Equal area projection of the stable end point directions (dots) and great circle trajectories identified from site PDA11. Solid (open) stars indicate the first (last) points identified on the great circle paths during thermal demagnetization, and lower (upper) hemisphere projection.

magnetic remanence (Figure 6). Subsequent AF demagnetization of the SIRM with a maximum alternating field of 200 mT generally removes 85-90% of the remanence, indicating that the principal remanence carriers are the lower-coercivity minerals, probably magnetite. Thermal demagnetization experiments of samples containing two orthogonal IRMs (1.5 T and 0.3 T) again demonstrate that the principal magnetic carrier is magnetite with unblocking temperatures between 500° and 580°C. In some samples, minor amounts of goethite are also present, as is indicated by a low unblocking temperature component (~100°C) with high coercivities (0.3-1.5 T). The occasional presence of pyrrhotite is also indicated from the IRM acquisition curves (saturation not reached until ~400 mT) and an inflection in the intensity decay curves at 300°-350°C. This, in combination with the paleomagnetic data, indicates magnetite as the main carrier of the B direction of magnetization, with minor and occasional amounts of goethite and pyrrhotite.

Thermomagnetic experiments of samples in which stable end point directions were identified yield a peak in heating at about 500°C (Figure 7). This probably results from the alteration of iron sulphides or goethite to magnetite, and explains the thermochemical alteration recognized during thermal demagnetization of the NRM. An interesting feature of the cooling curve from these samples is the distinct inflection at 300°C (Figure 7a). The significance of this is not yet clear, but it may indicate the production of an iron sulphide, probably pyrrhotite.

From samples that displayed great circle behavior during NRM demagnetization, however, thermochemical alteration during heating is less pronounced or nonexistent (Figure 7b), and thermomagnetic curves indicate the presence only of fine grained magnetite with Curie temperatures of 580°C, with no marked inflection in the cooling curve. As is shown in Figure 7, the thermomagnetic curves for these samples are not reversible, due to the production of magnetite at higher temperatures.

Despite the rather low magnetic content of these samples, it was possible in some cases to carry out hysteresis measurements, and the curves obtained are generally slightly wasp-waisted. In this case, this effect is thought to be caused by the presence of different magnetic minerals with contrasting coercivities (ie magnetite and pyrrhotite). However, further rock magnetic studies are required to verify this. Also, when the parameters Mrs/Ms and Hcr/Hc obtained from these experiments are displayed in the typical Day diagram [*Day et al.*, 1977], the data plot either in PSD field or outside the fields due to high Hcr/Hc ratios. This

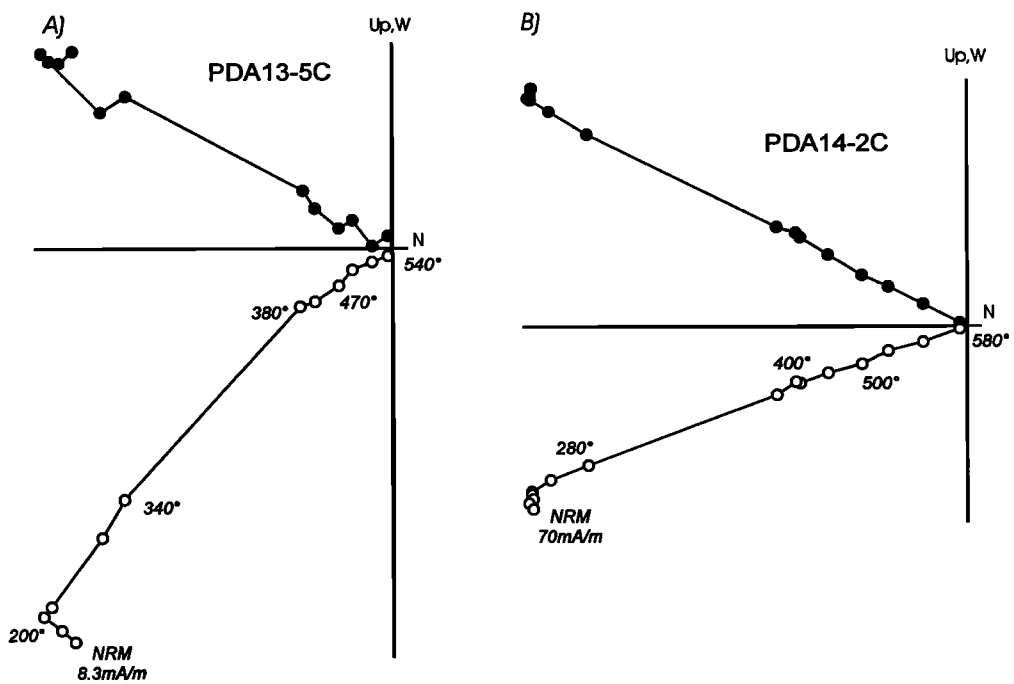


Figure 5. Thermal demagnetization characteristics of (a) samples collected from the sill and (b) immediately adjacent baked sediments. Notation is the same as for Figure 2.

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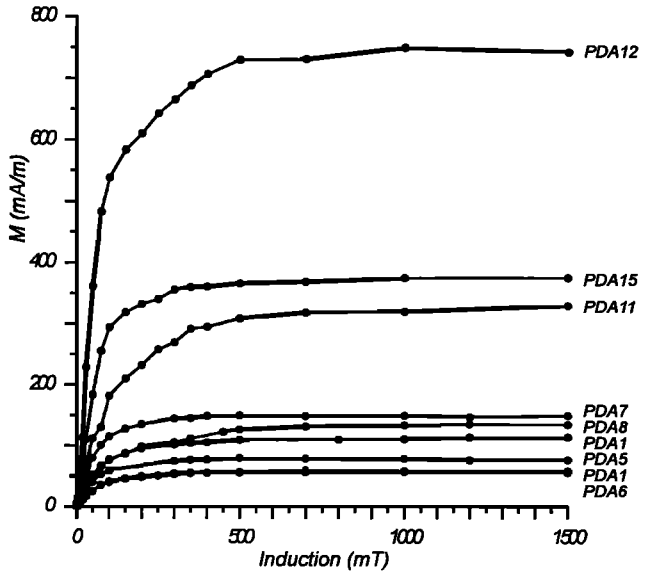


Figure 6. IRM acquisition curves.

again is typical for samples with mixed magneto-mineralogy, and demonstrates the dominance of a softer fraction as the magnetic carrier.

6. Discussion and Conclusions

One ancient direction of magnetization (component B) has been clearly identified in these Lower Devonian sediments from northern France. Unfortunately, it was not possible to carry out a meaningful fold test on the results owing to the relatively uniform structural attitude of the sediments. However, the sample and site mean directions (11 sites) are extremely well grouped (Figure 8) and are not similar to any younger directions of magnetization expected for this locality. Thus component B is considered to be primary in origin, and closely linked with the rock age. Supporting this interpretation is, firstly, the fact that samples from the lower part of the sequence also seem to record the inverse paleomagnetic direction, and, secondly, a positive contact test. The resulting paleopole of 315°E, 13°S (11 sites) corresponds to a paleolatitude of 19°S for northern France in the earliest Devonian (Figure 9), placing the Armorican Massif adjacent to the southern margin of Avalonia and Baltica. These are the first reliable paleomagnetic data from Early Devonian sequences of the Armorican Massif and clearly demonstrate that this sector of the Rheic Ocean had closed by this time. This is in agreement with geological evidence [*Franke et al.*, 1995], and the Early Devonian metamorphic event recognized in the Armorican Massif which is thought to indicate the onset of continent-continent collision [*Jones*, 1994].

These new data also have important implications for the geodynamic evolution of Variscan Europe and the Armorican Microplate concept. Comparing these results with those of *Tait et al.* [1994] from Upper Silurian rocks of the Barrandian Basin, Bohemian Massif, the inclination values obtained from both regions are in excellent agreement, but there are major differences in the paleomagnetic declinations. In conjunction with geological evidence, the Bohemian results indicate large scale anticlockwise rotation of the Bohemian Massif between latest Silurian and Late Devonian times. These rotations, however, are not recognzsed in the Early Devonian results from Armorican Massif. While it may be possible that the magnetizations in the Armorican Massif were acquired after rotation, this is considered unlikely as it would require extremely rapid rotation of the Armorican microplate, i.e. some 140° between the latest Silurian (Pridoli) and earliest Devonian (Lochkovian). The alternative and preferred interpretation is that these results indicate a major tectonic dislocation between these two regions which accommodated significant amounts of differential rotation. Thus while the two massifs were at similar paleolatitudes in the Early Devonian, as the faunal record also indicates, they did not form a coherent tectonic unit but instead belonged to two distinct microplates. The term "Armorican microplate", therefore, is no longer tenable, and the term Armorican terrane assemblage (or ATA) is proposed, with the implication that while the Gondwana-derived terranes of central and eastern Europe south of the Rhenohercynian zone (i.e., Avalonia) had similar drift histories in the early Paleozoic, they did not form a rigid microplate.

The question as to whether the Armorican Massif remained adjacent to the northern margin of Gondwana cannot be conclusively resolved by these results because of ambiguity within the paleomagnetic data set with regard to the apparent polar wander (APW) path of Gondwana. Details of this controversy are beyond the scope of this paper, but essentially two different models have

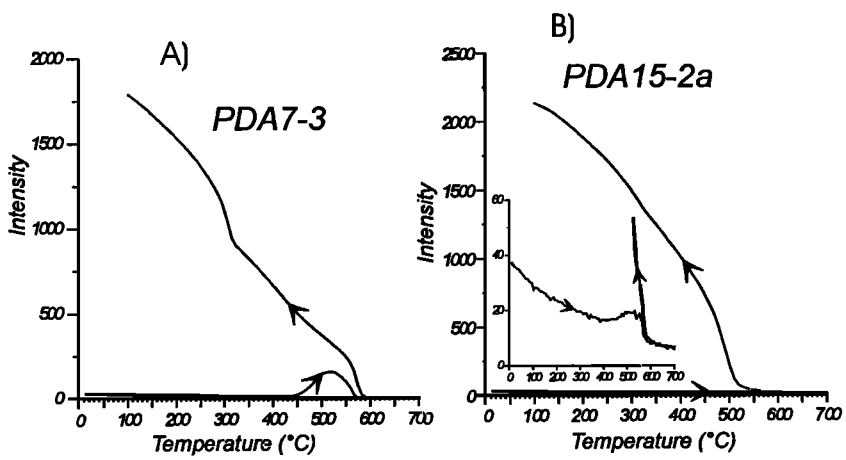


Figure 7. Thermomagnetic curves obtained from the Lower Devonian carbonates: (a) PDA7-3 (stable end point direction during NRM demagnetization), and (b) PDA15-2a (great circle behaviour during NRM demagnetization). The inset in Figure 7b shows a detail of the heating curve.

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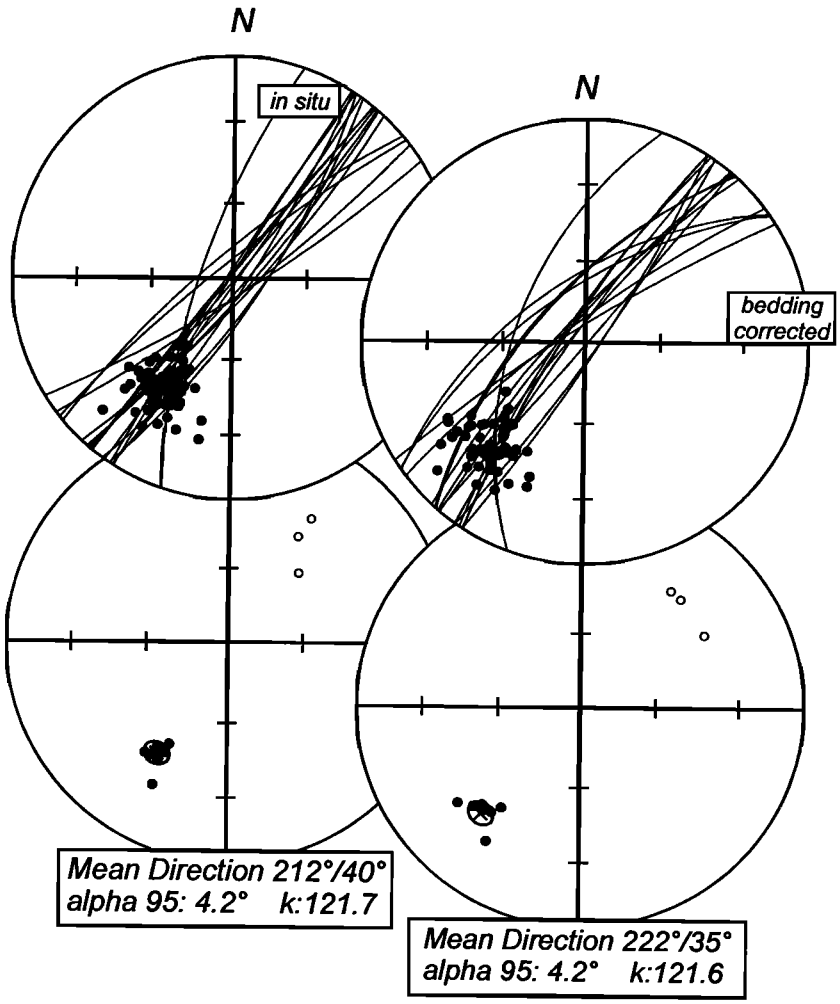


Figure 8. Equal area projection of the sample directions and site mean directions, both before and after bedding correction, obtained in this study. Notation is the same as for Figure 1.



been proposed, and the two different options for the position of Gondwana in the Early Devonian are shown in Figure 9. The more northerly position of Gondwana (dotted outline) is based on the more complex APW path and uses the paleomagnetic data of *Hargraves et al.* [1987] obtained from Early Devonian age [*Moreau et al.*, 1994] rocks of the Aïr igneous complex in Niger, north Africa. This model places the paleo-South Pole next to southern Chile, requires high drift rates for Gondwana, and places its northern margin in equatorial latitudes in the Early Devonian.

Figure 9. Paleogeographic reconstruction for Early Devonian times, using the data from the present study,as well as from *Torsvik et al.* [1992] for Baltica, *Channell et al.* [1992] for Avalonia, and *Smethurst et al.* [1998] for Siberia. Two different options for the position of Gondwana are also indicated. The dotted outline indicates the palaeoposition according to the more complex APW path based on the paleomagnetic data of *Hargraves et al.* [1987]. The solid outline indicates the preferred position using the more conservative APW path of *Bachtadse and Briden* [1991]. The star indicates the sampling locality in north west France.

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The alternative APW path proposed by *Bachtadse and Briden* [1991], however, keeps northern Africa at fairly high palaeolatitudes in the Early Devonian (Figure 9b, solid outline). This model, the current author's preferred option, is more in keeping with geological indicators, is the current author's preferred option, and implies an oceanic separation of some 3000 km between the southern margin of the Armorican terranes and northern Gondwana. For further arguments and discussion of the APW path for Gondwana the reader is referred to *Bachtadse et al.* [1995], *Van der Voo*, [1993], and references therein.

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